Same-depth inverse rigidity across two preceding record atoms

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For a positive odd integer x, put

$$U_+(x) := \frac{3x+1}{2^{v_2(3x+1)}}. \tag{1}$$

For an exponent word $w = (a_1, \dots, a_L)$, write

$$B(w) := \sum_{i=1}^{L} a_i, \qquad C(w) := \sum_{i=1}^{L} 3^{L-i} 2^{a_1+\cdots+a_{i-1}}. \tag{2}$$

Thus

$$2^{B(w)} U_+^L(x) = 3^L x + C(w). \tag{3}$$

On an interior deficit-four record tail, use the atom notation

$$A := 1, \qquad B := 12, \qquad C := 1113, \qquad D := 1122. \tag{4}$$

The length and excess mass of a nontrivial atom are

P	B	C	D
$\lvert P\rvert$	2	4	4
$B(P) - \lvert P\rvert$	1	2	2.

(5)

If t is an odd target and $M \geq L$, let $\mathcal{R}_L(t; M)$ denote the reverse exponent strings $(d_1, \dots, d_L)$ of total mass at most M which give a positive odd L-step predecessor of t. They are generated by

$$t_0 = t, \qquad t_i = \frac{2^{d_i} t_{i-1} - 1}{3}, \tag{6}$$

with every t_i a positive odd integer. For a recursion with ℓ steps still unprocessed, only $t_i \bmod 3^\ell$ is needed.

Fix $P, Q \in \{B, C, D\}$ and $r, s \geq 0$, and consider the exact word

$$W_{P,Q}(r,s) := PA^rQA^sC. \tag{7}$$

Put

$$\begin{aligned} L &:= |W_{P,Q}(r,s)|, & B_0 &:= B(W_{P,Q}(r,s)), \\ \delta_P &:= \begin{cases} 1, & P = C, \\ 0, & P \in \{B, D\}, \end{cases} & M &:= B_0 - \delta_P. \end{aligned} \tag{8}$$

Lemma 1 (finite reverse-profile certificate)**.** *Let q be a positive odd start for the exact word (7), and let $y = U_+^L(q)$. Then*

$$\boxed{\mathcal{R}_L(y; M) = \begin{cases} \{\mathrm{rev}\, W_{P,Q}(r,s)\}, & P \in \{B, D\}, \\ \{\mathrm{rev}(1121\, A^rQA^sC)\}, & P = C. \end{cases}} \tag{9}$$

In particular, the minimum inverse mass is B_0 for $P = B, D$ and $B_0 - 1$ for $P = C$, and the minimum-mass reverse word is unique.

Proof. The reverse sets satisfy the finite recursion

$$\mathcal{R}_{\ell+1}(t; N) = \bigcup_{\substack{d \geq 1,\ d \leq N-\ell \\ 3 \mid 2^d t - 1}} \left\{ (d, \gamma) : \gamma \in \mathcal{R}_\ell\left(\frac{2^d t - 1}{3}; N - d \right) \right\}. \tag{10}$$

The excess $M - L$ is at most 6. Hence every exponent tested in (10) is at most 7.

Process the last $L - |P|$ reverse steps and retain only branches leaving at least one unit of mass per step of P. For each retained branch, record

$$(u, h) := \left(t_{L-|P|} \bmod 3^{|P|}, M - \sum_{i=1}^{L-|P|} d_i \right). \tag{11}$$

The complete profiles are as follows; every displayed pair has multiplicity one.

(P,Q)	profile $\Pi_{P,Q}(r,s)$
(B,B)	$\{(4,3),(0,2)\}$
(B,C)	$\{(6,4),(7,2),(4,3)\}$
(B,D)	$\{(4,3),(0,2)\}$
(C,B)	$\{(20,5),(41,4)\}$
(C,C)	$\{(50,6),(39,4),(20,5)\}$
(C,D)	$\{(20,5),(41,4)\}$
(D,B)	$\{(10,6),(41,4),(21,5)\}$
(D,C)	$\{(45,7),(19,5),(10,6),(41,4)\}$
(D,D)	$\{(43,4),(10,6),(41,4),(21,5)\}$.

(12)

When $(P,Q)=(D,C)$ and $r=0$, there is one additional pair $(39,4)$.

Here is an exact induction certificate for (12). At $r=s=0$, the triples $(L,M,y \bmod 3^L)$ are

PQ	BB	BC	BD
(L,M,y_0)	$(8,12,2045)$	$(10,15,33959)$	$(10,15,51374)$

(13)

PQ	CB	CC	CD
(L,M,y_0)	$(10,14,10064)$	$(12,17,375131)$	$(12,17,274448)$

(14)

PQ	DB	DC	DD
(L,M,y_0)	$(10,15,37037)$	$(12,18,73325)$	$(12,18,504083)$.

(15)

Starting from these residues, substitution of $1 \le d \le 7$ in (10) gives precisely the nine rows of (12). Inserting one additional letter A in either run raises both L and M by one. More explicitly, if $w=UV$, $b=B(U)$, and $k=|V|$, then

$$C(UAV) = 3^{k+1}C(U) + 2^b\big(3^k + 2C(V)\big). \tag{16}$$

Substitution of this identity in (10) leaves every profile in (12) unchanged under $r \mapsto r+1$ and $s \mapsto s+1$, with one exception. Writing $\Pi^*_{D,C}$ for the four-pair set displayed in the (D,C) row, that exception is

$$\begin{aligned} \Pi_{D,C}(0,s) &= \Pi^*_{D,C} \cup \{(39,4)\}, \\ \Pi_{D,C}(r,s) &= \Pi^*_{D,C} \qquad (r \ge 1). \end{aligned} \tag{17}$$

The s-lift preserves both alternatives in (17), while the first r-lift deletes $(39,4)$. Equations (13)–(17) are a two-variable induction proving (12). This

certificate involves only division by 3, reduction modulo 9 or 81 at the terminal boundary, and the seven possible values of d; there is no unlisted asymptotic or positivity step.

It remains to complete the first atom. Direct use of (10) gives

P	nonempty completion among all pairs in (12)
B	$\mathcal{R}_2(4;3) = \{(2,1)\}$,
C	$\mathcal{R}_4(20;5) = \{(1,2,1,1)\}$,
D	$\mathcal{R}_4(10;6) = \{(2,2,1,1)\}$.

(18)

All other completions are empty; explicitly,

$$\begin{aligned} &\mathcal{R}_2(0;2) = \mathcal{R}_2(6;4) = \mathcal{R}_2(7;2) = \varnothing, \\ &\mathcal{R}_4(41;4) = \mathcal{R}_4(50;6) = \mathcal{R}_4(39;4) = \varnothing, \\ &\mathcal{R}_4(21;5) = \mathcal{R}_4(45;7) = \mathcal{R}_4(19;5) = \mathcal{R}_4(43;4) = \varnothing. \end{aligned} \tag{19}$$

In (18)–(19), a residue denotes any odd target in that residue class modulo $3^{|P|}$; recursion (10) depends on no further information.

The unique retained suffix leading to $(4,3)$ or $(10,6)$ is the reverse of A^rQA^sC, and (18) appends the reverse of B or D. The unique suffix leading to $(20,5)$ is again the actual suffix, while its completion $(1,2,1,1)$ is the reverse of 1121. This proves (9). □

Theorem 2 (two-preceding-atom same-depth minimum)**.** *Let q be the start and y the endpoint of the exact word $W_{P,Q}(r,s)$ in (7). Among all positive odd L-step predecessors of y, the unique least one is*

$$\boxed{q_{\min} = \begin{cases} q, & P \in \{B, D\}, \\ (q-1)/2, & P = C. \end{cases}} \tag{20}$$

When $P = C$, the second line is exactly the already known half-coalescing seed at the first atom C; its word is $1121A^rQA^sC$.

Proof. Lemma 1 gives the minimum inverse mass M and its unique word. For $P = C$, the exact local identity

$$U_+^4\left(\frac{q-1}{2}\right) = U_+^4(q) \tag{21}$$

replaces the first word 1113 by 1121, proving that the start in (20) is attached to that minimum word.

We first record coarse lower bounds on this start. Exactness of the first atom gives

$$\begin{aligned} q &\equiv 11 \pmod{16} \quad (P = B), \\ q &\equiv 79 \pmod{128} \quad (P = C), \\ q &\equiv 71 \pmod{128} \quad (P = D). \end{aligned} \tag{22}$$

Every suffix A^rQA^sC starts with exponent 1. Direct iteration of the finitely many small representatives gives

P	q	initial exponent or atom pattern
B	11	123
	27	BA^2DB
	43	122
	59	1214
C	79	11134
D	71	$DBAB$.

(23)

None has the form PA^rQA^sC. Therefore

$$q > 64 \quad (P = B), \qquad \frac{q-1}{2} > 64 \quad (P = C), \qquad q > 128 \quad (P = D). \tag{24}$$

Let q_* denote the value in (20), and let C_* be the carry of its minimum word. From (5) and (8), the bounds in (24) imply

$$\boxed{q_* > 2^{M+1-L}.} \tag{25}$$

Indeed, the right side is at most 64 for $P = B, C$ and at most 128 for $P = D$.

Now let $v = (c_1, \dots, c_L)$ be any other exact word from a positive odd z to y, and put $B' = B(v)$. Lemma 1 gives $B' \geq M + 1$. Positivity of the exponents gives the uniform carry estimate

$$C(v) \leq 2^{B'-L}(3^L - 2^L) < 2^{B'-L}3^L. \tag{26}$$

The minimum affine equation and (25) imply $y > (3/2)^L$. Hence

$$\begin{aligned} z &= \frac{2^{B'}y - C(v)}{3^L} \\ &> \frac{2^{B'}y}{3^L} - 2^{B'-L} \\ &\geq \frac{2^{M+1}y}{3^L} - 2^{M+1-L} \\ &= 2q_* + \frac{2C_*}{3^L} - 2^{M+1-L} > q_*. \end{aligned} \tag{27}$$

Thus the minimum-mass start is also the unique least positive start. □

Corollary 3 (two preceding atoms give no new same-depth descent)**.** *Suppose an interior record tail contains*

$$PA^rQA^sC, \qquad P, Q \in \{B, C, D\}. \tag{28}$$

If $P = B$ or D, no same-depth inverse branch starts below the actual block start. If $P = C$, the unique least branch is only the local half-seed $(q-1)/2$ belonging to that earlier C atom. Consequently a new pullback of the terminal C half-coalescence cannot produce a same-depth descent while crossing only its two preceding nontrivial atoms. Such a construction must change raw inverse depth or cross at least three preceding nontrivial atoms.

Proof. Apply Theorem 2. Its $P = C$ alternative is (20), so it introduces no inverse mechanism beyond the already classified local C half-seed. □